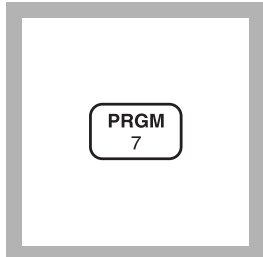


NITROGEN, AMMONIA, High Range, Test 'N Tube

(0 to 50 mg/L $\text{NH}_3\text{-N}$)

For water, wastewater, and seawater

Salicylate Method*

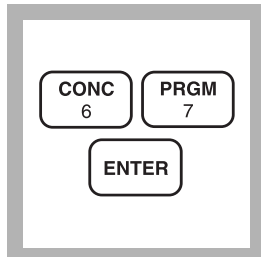


1. Enter the stored program number for nitrogen, ammonia, high range Test 'N Tube ($\text{NH}_3\text{-N}$) method.

Press: **PRGM**

The display will show:

PRGM ?

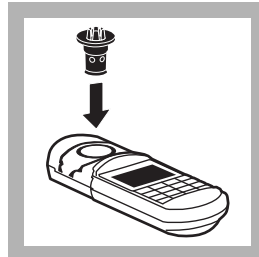


2. Press: **67 ENTER**

The display will show **mg/L, $\text{NH}_3\text{-N}$** and the **ZERO** icon.

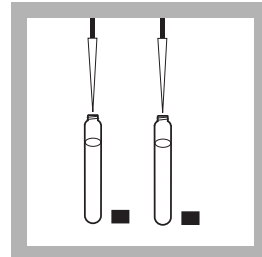
Note: For alternate forms (NH_3), press the **CONC** key.

Note: For proof of accuracy, use a 10-mg/L nitrogen, ammonia standard in place of the sample.

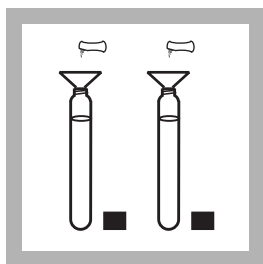


3. Insert the COD/TNT Adapter into the cell holder by rotating the adapter until it drops into place. Then push down to fully insert it.

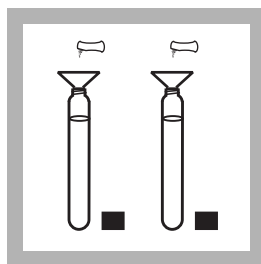
Note: For increased performance, a diffuser band covers the light path holes on the adapter. Do not remove the diffuser band.



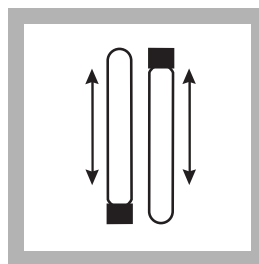
4. Remove the caps from 2 AmVer Diluent Reagent High Range Vials. Add 0.1 mL of sample to one vial (the sample). Add 0.1 mL of deionized water to the other (the blank).



5. Add the contents of 1 Ammonia Salicylate Reagent Powder Pillow for 5 mL Sample to each vial.

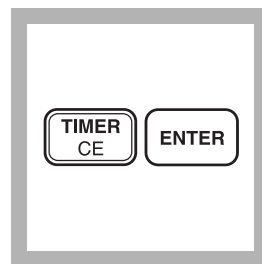


6. Add the contents of 1 Ammonia Cyanurate Reagent Powder Pillow for 5 mL Sample to each vial.



7. Cap the vials tightly and shake thoroughly to dissolve the powder.

Note: A green color will develop if ammonia is present.



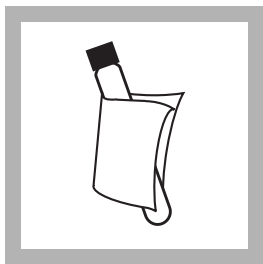
8. Press:

TIMER ENTER

A 20-minute reaction period will begin.

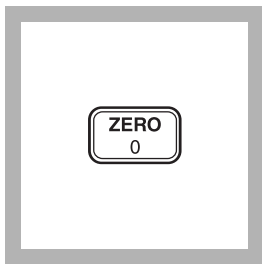
* Adapted from *Clin. Chim. Acta*, **14** 403 (1966).

NITROGEN, AMMONIA, High Range, Test 'N Tube, continued

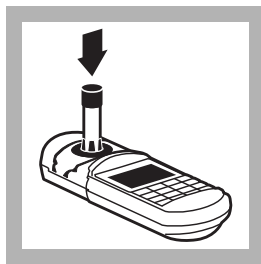


9. Clean the outside of the vial with a towel. After the timer beeps, place the blank into the vial adapter. Tightly cover the vial with the instrument cap.

***Note:** Wipe with a damp cloth and follow with a dry one to remove fingerprints and other marks.*

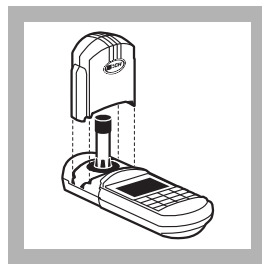


10. Press: **ZERO**
The cursor will move to the right, then the display will show:
0 mg/L NH₃-N

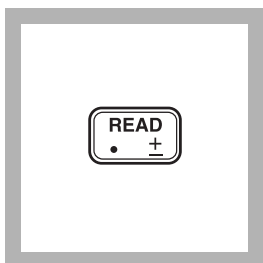


11. Place the prepared sample in the adapter. Push straight down on the top of the vial until it seats solidly into the adapter.

***Note:** Do not move the vial from side to side as this can cause errors.*



12. Tightly cover the vial with the instrument cap.



13. Press: **READ**

The cursor will move to the right, then the result in mg/L NH₃-N will be displayed.

***Note:** Standard Adjust may be performed using a prepared standard (see Standard Adjust in Section 1).*

NITROGEN, AMMONIA, High Range, Test 'N Tube, continued

Sampling and Storage

Collect samples in clean plastic or glass bottles. Best results are obtained with immediate analysis. If chlorine is known to be present, add one drop of 0.1 N sodium thiosulfate for each 0.3 mg/L Cl_2 in a one liter sample. Preserve the sample by reducing the pH to 2 or less with hydrochloric acid (at least 2 mL). Store at 4 °C (39 °F) or less. Preserved samples may be stored up to 28 days. Before analysis, warm samples to room temperature and neutralize with 5.0 N sodium hydroxide. Correct the test result for volume additions.

Accuracy Check

Standard Additions Method

- a) Snap the top off an Ammonia PourRite Ampule Standard, 150 mg/L $\text{NH}_3\text{-N}$.
- b) Use the TenSette Pipet to add 0.2, 0.4 and 0.6 mL of standard to three 25-mL samples. Swirl to mix.
- c) Analyze each sample as described above. The ammonia concentration should increase approximately 1.2 mg/L $\text{NH}_3\text{-N}$ for each 0.2 mL of standard added.
- d) If these increases do not occur, see *Standard Additions* in *Section 1* for more information.

Standard Solution Method

To check accuracy, use a 10 or 50 mg/L Nitrogen, Ammonia Standard Solution or use a Nitrogen, Ammonia Voluette Ampule Standard, 50 mg/L.

Method Performance

Precision

In a single laboratory, using a standard solution of 50 mg/L ammonia nitrogen ($\text{NH}_3\text{-N}$) and two representative lots of reagent with the instrument, a single operator obtained a standard deviation of ± 5 mg/L $\text{NH}_3\text{-N}$.

Estimated Detection Limit

The estimated detection limit for program 67 is 1 mg/L $\text{NH}_3\text{-N}$. For more information on the estimated detection limit, see *Section 1*.

NITROGEN, AMMONIA, High Range, Test 'N Tube, continued

Interferences

The following ions may interfere when present in concentrations exceeding those listed below.

In some lab environments, airborne cross contamination of the blank is possible. Complete preparation of the blank before opening or handling any samples or standards to avoid transfer of ammonia. If sample or standard containers have already been open, move to a separate area of the lab to prepare the blank.

Substance	Concentration and Suggested Treatments
Acidic or basic samples	Adjust to approximately pH 7. Use 1 N Sodium Hydroxide Standard Solution for acidic samples and 1 N Hydrochloric Acid Standard Solution for basic samples.
Calcium	50,000 mg/L as CaCO_3
Glycine, hydrazine	Will cause intensified colors in the prepared sample.
Magnesium	300,000 mg/L as CaCO_3
Iron	Eliminate iron interference as follows: 1. Determine the amount of iron present in the sample using one of the total iron procedures. 2. Add the same iron concentration to the deionized water in step 4. 3. The interference will then be successfully blanked out.
Nitrite	600 mg/L as NO_2^- -N
Nitrate	5,000 mg/L as NO_3^- -N
Orthophosphate	5,000 mg/L as PO_4^{3-} -P
Sulfate	6,000 mg/L as SO_4^{2-}
Sulfide	Sulfide will intensify the color. Eliminate sulfide interference as follows: 1. Measure about 350 mL of sample in a 500 mL Erlenmeyer flask. 2. Add the contents of one Sulfide Inhibitor Reagent Powder Pillow. Swirl to mix. 3. Filter the sample through folded filter paper. Use the filtered solution in step 4.
Turbidity and color	Give erroneous high values. Samples with severe interferences require distillation. Hach recommends the distillation procedure using the Hach General Purpose Distillation Set.

NITROGEN, AMMONIA, High Range, Test 'N Tube, continued

Summary of Method

Ammonia compounds combine with chlorine to form monochloramine. Monochloramine reacts with salicylate to form 5-aminosalicylate. The 5-aminosalicylate is oxidized in the presence of a sodium nitroprusside catalyst to form a blue-colored compound. The blue color is masked by the yellow color from the excess reagent present to give a green-colored solution.

Safety

Good safety habits and laboratory techniques should be used throughout the procedure. Consult the *Material Safety Data Sheets* for information specific to the reagents used. For additional information, refer to *Section 3*.

Pollution Prevention And Waste Management

The ammonia salicylate reagent contains sodium nitroferricyanide. Cyanide solutions are regulated as hazardous wastes by the Federal RCRA. Collect cyanide solutions for disposal as reactive (D001) waste. Be sure cyanide solutions are stored in a caustic solution with pH >11 to prevent release of hydrogen cyanide gas. See *Section 3* for further information in proper disposal of these materials.

REQUIRED REAGENTS

AmVerTM Reagent Set for Nitrogen, Ammonia, High Range, TNT (25 tests)26069-45
Includes: (1) 23952-66, (1) 23954-66, (1) 272-42, * (50) AmVer HR Vials

Description	Quantity Required		Unit	Cat. No.
	Per Test			
AmVer TM HR Reagent Test 'N Tube TM Vials	2 vials	50/pkg		*
Ammonia Salicylate Reagent Powder Pillows.....	2 pillows	50/pkg		23952-66
Ammonia Cyanurate Reagent Powder Pillows	2 pillows	50/pkg		23954-66

REQUIRED APPARATUS

COD/TNT Adapter.....	1	each	48464-00
Pipet, TenSette [®] , 0-1 mL	1	each	19700-01
Pipet Tips for 19700-01.....	varies	50/pkg	21856-96
Test Tube Rack.....	1-3	each	18641-00
Funnel, micro (for reagent addition)	1	each	25843-35

* Not available separately.

NITROGEN, AMMONIA, High Range, Test 'N Tube, continued

OPTIONAL REAGENTS

Description	Quantity Required		Cat. No.
	Per Test	Unit	
Nitrogen, Ammonia Standard Solution, 50 mg/L NH ₃ -N.....		500 mL.....	14791-50
Nitrogen, Ammonia Standard Solution, 10 mg/L NH ₃ -N.....		500 mL.....	153-49
Ammonia Standard Solution, PourRite™ ampules, 150 mg/L NH ₃ -N, 2 mL.....		20/pkg.....	21284-20
Hydrochloric Acid, ACS.....		500 mL.....	134-49
Sodium Hydroxide Standard Solution, 5.0 N		50 mL.....	2450-26
Sodium Hydroxide Standard Solution, 1.0 N		100 mL.....	1045-32
Sodium Thiosulfate Standard Solution, 0.1 N		100 mL.....	323-32
Sulfide Inhibitor Powder Pillows.....		100/pkg.....	2418-99
Sulfuric Acid, 1.00 N.....		100 mL MDB.....	1270-32
Wastewater Influent Standard, Inorganic (NH ₃ -N, NO ₃ , PO ₄ , COD, SO ₄ , TOC).....		500 mL.....	28331-49
Water, deionized.....		4 L	272-56

OPTIONAL APPARATUS

Cylinder, 25 mL, graduated, mixing.....	each.....	20886-40
Distillation Apparatus Set, general purpose	each.....	22653-00
Heater and Support Apparatus (for distillation), 115 VAC.....	each.....	22744-00
Heater and Support Apparatus (for distillation), 230 VAC	each.....	22744-02
Filter Paper, folded.....	100/pkg.....	1894-57
Flask, Erlenmeyer, 500 mL.....	each.....	505-49
Funnel, analytical (for filtering).....	each.....	1083-68
Jack, laboratory (use with distillation apparatus)	each.....	22743-00
pH Indicator Paper, 1 to 11 pH	5 rolls/pkg.....	391-33
Pipet Tips, for 19700-01 TenSette Pipet	1000/pkg.....	21856-28
PourRite™ Ampule Breaker	each.....	24846-00
Sample Cell, 10-20-25 mL, w/cap	6/pkg.....	24019-06

For Technical Assistance, Price and Ordering

In the U.S.A.—Call 800-227-4224

Outside the U.S.A.—Contact the Hach office or distributor serving you.